

School: Computer Science
Institution: University of Windsor
Term: Winter 2026
Course: Comp-3150 (03-60-315-1) : Database Management Systems
Instructor: Dr. C. I. Ezeife
Assignment #3 Solution: Total: 50 marks
Handed Out: Thurs. Feb. 26, 2026; **Due:** Thurs. Mar. 19, 2026

Objective of Assignment: To test on knowledge and use of relational database query languages SQL and relational algebra for implementing relational databases.

Scope: Assignment covers materials from Chapters 6, 7 and 8 of book discussed in class.

Electronic Assignment Submission: Done through <http://brightspace.uwindsor.ca>

Marking Scheme : The mark for each of the questions is indicated beside each question.

Academic Integrity Statement: Remember to submit only work that is yours and include the following confidentiality agreement and statement at the beginning of your assignment.

CONFIDENTIALITY AGREEMENT & STATEMENT OF HONESTY

I confirm that I will keep the content of this assignment/examination confidential.

I confirm that I have not received any unauthorized assistance in preparing for or doing this assignment/examination. I confirm knowing that a mark of 0 may be assigned for copied work.

Student Signature

Student Name (please print)

Student I.D. Number

Date

Marking Scheme : The mark for each question and sub question is shown with the question below. Place your solutions in tables where possible.

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Question	Mark
1	/15
2	/10
3	/5
4	/5
5	/5
6	/10
Total	/50

CHAPTER 6: Basic SQX

1. Given a database state of of the Farmer-growscrops-inRegions database shown in Figure 1.1, with schema shown in Figure 1.2, answer question (i) to (v) given below.

(Total for que 1 is 15 marks)

Note : Ssn, Name, Age, Regionid are the social security number, name, age and Regionid of the farmer respectively. Also, Regname, Cropbudget and managerid represent the regionid (e.g., SWO for south western Ontario), all crop budget (amount) for growing in this region (e.g., 100 for \$100M), and managerid (a farmer who oversees farmer affairs in this region). Revenue from grown crop can be measured in millions of dollars as well (e.g., 0.5 means half a million). A farmer may grow crop in multiple regions.

**** Note that in creating these tables in the same Oracle database space in your account, you need to give them unique names not already given to any tables in your database. For example, you may need to call table Farmer something like Farmer_asn3 if you already have a table called Farmer from your assignment 2. Rename all the other tables in a similar way.**

Fig 1.1: An Example Database State of the Farmer-growscrops-inRegions Database

Farmer

<u>Ssn</u>	Name	Age	Regionid
10	Jobe Bata	65	1
20	Monica Kap	80	1
30	Peter Good	22	2
40	Kate Lee	47	3
50	Ted Tam	50	4

Region

<u>Regionid</u>	Regname	Cropbudget	managerid
1	SWO	5.00	10
2	EO	4.00	10
3	SC	9.00	50
4	SEO	6.00	50

RContact

<u>Regionid</u>	Fullname	RPHONE
1	Southwest Ontario	NULL
2	Eastern Ontario	1(733)524-0100
3	Southcentral Ontario	1(866)663-3161
4	Southeastern Ontario	1(877)633-2000

Growscrops

<u>Ssn</u>	<u>Regionid</u>	<u>crop</u>	Revenue
10	1	corn	2.50
10	1	carrots	4.00
20	1	potatoes	9.00
30	2	potatoes	11.00
40	3	peas	15.30
50	4	potatoes	10.30
50	3	peas	14.00

**

Fig 1.2: Schema of Farmer database of Figure 1.1

Farmer

<u>Ssn</u>	Name	Age	Regionid
------------	------	-----	----------

Region

<u>Regionid</u>	Regname	Cropbudget	managerid
-----------------	---------	------------	-----------

RContact

<u>Regionid</u>	Fullname	RPHONE
-----------------	----------	--------

Growscrops

<u>Ssn</u>	<u>Regionid</u>	<u>crop</u>	Revenue
------------	-----------------	-------------	---------

- List all the referential integrity constraints that should hold on the database schema
(2.5 marks)
- Write appropriate SQL DDL statements to define the database with the integrity constraints and store in a text file called userid_farmerschema.sql. Attach this file or also show it in your script file of (v) using more file.sql command before or after running Sqlplus. Do the same for the files in (iii) and (iv).
(2.5 marks)
- To insert the data in the database tables, also write appropriate SQL DML instructions in a text file called userid_farmerdata.sql.
(2.5 marks)

- iv. To remove any inserted data and destroy all created tables in the farmer database, write appropriate SQL DML and DDL statements in a text file called `userid_farmerdroptable.sql` to first delete all data in the tables and then drop the tables. (2.5 marks)
- v. Using Oracle Sqlplus, implement this database design by creating all the tables with the integrity constraints using the SQL DDL you defined in (ii) above. You can create all these SQL DDL for creating the 4 tables by running your .sql file at the SQL prompt with the command:
`@userid_farmerschema.sql`. After creating your tables successfully, you load your data with the .sql file you created in (iii) above by running `@userid_farmerdata.sql`. If there are errors and you need to correct them, you might want to delete the tuples and drop the tables first using the .sql file you created in (iv) above as with `@userid_farmerdroptable.sql` before re-creating the schema and re-loading the data. Then, using a script file, show the contents of all 4 tables in the database by running: `select * from each of the tables` and saving on script file called `username_assn3que1.txt`. You can do this using the following sequence of Unix/Linux commands after you have created the database and inserted data. (5 marks)

(Note: remember to create the entity tables with primary keys before the relationship tables that reference them through foreign key attributes. When inserting data, do the same. If you need to delete the data and tables at any time, go in the reverse order (that is, delete the tuples that reference a primary key attribute tuple in another table, before deleting the parent primary keyed tuple))

```
>script username_assn3que1.txt
>sqlplus <username>
>password
sqlplus> select * from Farmer; //repeat this instruction for each table
sqlplus> exit //to exit sqlplus
exit // to exit and create script file
```

**Now attach the saved unix log file of your session that is in `username_assn3que1.txt` with an inclusion in this script file of all the 4 .sql files in questions (ii), (iii) and (iv) or the attachment of those files as your solution showing where you executed those files in Sqlplus.

Solution 1 (i) (mark: 2.5)

We will write a referential integrity constraint as $R.A \rightarrow S$ (or $R.(X) \rightarrow T$) whenever attribute A (or the set of attributes X) of relation R form a foreign key that references the primary key of relation S (or T).

Growscrops.SSN $\rightarrow$ FARMER
Growscrops.Regionid $\rightarrow$ REGION
Farmer.Regionid $\rightarrow$ REGION
Region.managerid $\rightarrow$ FARMER
RContact.Regionid $\rightarrow$ REGION

Solution 1 (ii): (mark: 2.5)

ONE POSSIBLE SET OF CREATE TABLE STATEMENTS TO DEFINE THE DATABASE IS GIVEN BELOW GIVEN IN THE FILE USERID_FARMERSHEMA.SQL IS:

```
-- This version of farmer grows crop database is for fall 2023 Assn3
CREATE TABLE FARMER_ASN3(
SSN NUMBER(3) NOT NULL,
NAME    VARCHAR2(15),
AGE NUMBER(3),
REGIONID  NUMBER(3),
PRIMARY KEY(SSN));

CREATE TABLE REGION_ASN3
(
REGIONID  NUMBER(3) NOT NULL,
REGNAME VARCHAR2(15) NOT NULL,
CROPBUDGET  NUMBER(10,2),
MANAGERID NUMBER(3),
PRIMARY KEY (REGIONID),
FOREIGN KEY(MANAGERID) REFERENCES FARMER_ASN3(SSN));
COMMIT;

CREATE TABLE RContact
(
REGIONID  NUMBER(3) NOT NULL,
FULLNAME VARCHAR2(30) NOT NULL,
RPHONE    VARCHAR2(20),
PRIMARY KEY (REGIONID),
FOREIGN KEY(REGIONID) REFERENCES REGION_ASN3(REGIONID));

CREATE TABLE GROWSCROPS_ASN3
(
SSN  NUMBER(3) NOT NULL,
REGIONID  NUMBER(3) NOT NULL,
CROP VARCHAR2 (15) NOT NULL,
REVENUE NUMBER(10, 2),
PRIMARY KEY(SSN, REGIONID, CROP),
FOREIGN KEY(SSN) REFERENCES FARMER_Asn3(SSN),
FOREIGN KEY(REGIONID) REFERENCES REGION_ASN3(REGIONID));

COMMIT;
```

Solution 1 (iii): (mark: 2.5)

One possible set of INSERT INTO TABLE statements to define the database is given below given in the file userid_farmerdata.sql is:

```
-- Now inserting data into the table FARMER_ASN3
```

```
INSERT INTO FARMER_ASN3
VALUES (10, 'Jobe Bata', 65, 1);

COMMIT;

INSERT INTO FARMER_ASN3
VALUES (20, 'Monica Kap', 80, 1);

COMMIT;

INSERT INTO FARMER_ASN3
VALUES (30, 'Peter Good', 22, 2);

COMMIT;

INSERT INTO FARMER_ASN3
VALUES (40, 'Kate Lee', 47, 3);

COMMIT;

INSERT INTO FARMER_ASN3
VALUES (50, 'Ted Tam', 50, 4);

COMMIT;

-- Now inserting data into the table REGION_ASN3

INSERT INTO REGION_ASN3
VALUES (1, 'SWO', 5.00, 10);

COMMIT;

INSERT INTO REGION_ASN3
VALUES (2, 'EO', 4.00, 10);

COMMIT;

INSERT INTO REGION_ASN3
VALUES (3, 'SC', 9.00, 50);

COMMIT;

INSERT INTO REGION_ASN3
VALUES (4, 'SEO', 6.00, 50);

COMMIT;

-- Data for RCONTACT starts next
INSERT INTO RCONTACT
VALUES (1, 'Southwest Ontario', 'NULL');
COMMIT;
```

```
INSERT INTO RCONTACT
VALUES (2, 'Eastern Ontario', '1(733)524-0100');
COMMIT;

INSERT INTO RCONTACT
VALUES (3, 'Southcentral Ontario', '1(866)663 3161');
COMMIT;

INSERT INTO RCONTACT
VALUES (4, 'Southeastern Ontario', '1(877)633-2000');
COMMIT
-- Data for GROWSCROPS_ASN3 starts next

INSERT INTO GROWSCROPS_ASN3
VALUES (10, 1, 'corn', 2.50);

COMMIT;

INSERT INTO GROWSCROPS_ASN3
VALUES (10, 1, 'carrots', 4.00);

COMMIT;

INSERT INTO GROWSCROPS_ASN3
VALUES (20, 1, 'potatoes', 9.00);

COMMIT;

INSERT INTO GROWSCROPS_ASN3
VALUES (30, 2, 'potatoes', 11.00);

COMMIT;

INSERT INTO GROWSCROPS_ASN3
VALUES (40, 3, 'peas', 15.30);
COMMIT;

INSERT INTO GROWSCROPS_ASN3
VALUES (50, 4, 'potatoes', 10.30);
COMMIT;

INSERT INTO GROWSCROPS_ASN3
VALUES (50, 3, 'peas', 14.00);
COMMIT;
```

Solution 1 (iv): (mark: 2.5)

One possible set of DELETE FROM TABLE statements and DROP TABLE statements to delete data from the database and drop the tables is given below given in the file `userid_farmerdroptable.sql` is:

```
delete from GROWSCROPS_ASN3;

commit;

delete from RCONTACT;
commit;

delete from REGION_ASN3;
commit;

delete from FARMER_ASN3;
commit;


drop table GROWSCROPS_ASN3 cascade constraints;
drop table RCONTACT cascade constraints;
drop table REGION_ASN3 cascade constraints;
drop table FARMER_ASN3 cascade constraints;

commit;
```

1 (v). (5 marks) for the script file showing correct interaction with Oracle Sqlplus creating and loading data in these 5 tables.

2. Specify the following 5 queries in SQL on the Farmercrop record database schema of Figure 1.1. (Total for que 2 is 10 marks)

- i. List all your 5 queries in the table below first in SQL. (5 marks)
- ii. Implement the answering of your 5 queries in 2(i) using Sqlplus and the same database you created in question 1, providing your execution and answers to these questions in a script file called `username_assn3que2.txt`. (5 marks)

- (a) Retrieve the full names and phone numbers of all regions where potatoes are grown.
- (b) Retrieve the Regname, Fullname, Rphone of all regions with total revenue of more than \$10M.
- (c) For each region, retrieve its regionid, number of farmers growing in that region, , and number of crops it grows and total revenue in that region.
- (d) Retrieve for each farmer, the farmer's name, and the farmer's crop growing region names.
- (e) Retrieve the crop name, regionid and regname of all crops that are not grown in more than one region.

Solution 2 (i): Queries(5 marks) and 2(ii) Results (5 marks)

- (a) Retrieve the full names and phone numbers of all regions where potatoes are grown.
`SELECT RC.FULLNAME, RC.RPHONE`


```
FROM RCONTACT RC, GROWSCROPS G
WHERE G.CROP = 'potatoes'
AND RC.REGIONID = G.REGIONID;
```

```
/*
```

FULLNAME	RPHONE
-----	-----
Southwest Ontario	NULL
Eastern Ontario	1 (733) 524-0100
Southeastern Ontario	1 (877) 633-2000

```
*/
```

-- (b) Retrieve the Regname, Fullname, Rphone of all regions with total revenue of more than \$10M.

```
SELECT R.REGNAME, RC.FULLNAME, RC.RPHONE, SUM(G.REVENUE)
FROM REGION R, RCONTACT RC, GROWSCROPS G
WHERE R.REGIONID = RC.REGIONID
AND RC.REGIONID = G.REGIONID
GROUP BY R.REGNAME, RC.FULLNAME, RC.RPHONE
HAVING SUM(G.REVENUE) > 10;
```

```
/*
```

REGNAME	FULLNAME	RPHONE
-----	-----	-----
SUM(G.REVENUE)		

SWO	Southwest Ontario	NULL
15.5		
SC	Southcentral Ontario	1 (866) 663 3161
29.3		
EO	Eastern Ontario	1 (733) 524-0100
11		

REGNAME	FULLNAME	RPHONE
-----	-----	-----
SUM(G.REVENUE)		

SEO	Southeastern Ontario	1 (877) 633-2000
10.3		

```
*/
```

-- (c) For each region, retrieve its regionid, number of farmers growing in that region, , and number of crops it grows and total revenue in that region.

```
SELECT G.REGIONID, COUNT(G.SSN), count(G.CROP), SUM(G.REVENUE)
FROM GROWSCROPS G
GROUP BY G.REGIONID;
```

/*

REGIONID	COUNT(G.SSN)	COUNT(G.CROP)	SUM(G.REVENUE)
1	3	3	15.5
2	1	1	11
4	1	1	10.3
3	2	2	29.3

*/

-- (d) Retrieve for each farmer, the farmer's name, and the farmer's crop growing region names.

```
SELECT DISTINCT F.NAME, R.REGNAME
FROM REGION R, FARMER F, GROWSCROPS G
WHERE F.SSN = G.SSN
AND R.REGIONID = G.REGIONID;
```

/*

NAME	REGNAME
Ted Tam	SEO
Monica Kap	SWO
Peter Good	EO
Jobe Bata	SWO
Kate Lee	SC
Ted Tam	SC

6 rows selected.

*/

-- (e) Retrieve the crop name, regionid and regname of all crops that are not grown in more than one region.

```
SELECT G.CROP, G.REGIONID, R.REGNAME
FROM GROWSCROPS G, REGION R
```

```

WHERE R.REGIONID = G.REGIONID
AND G.CROP NOT IN (
    SELECT G.CROP
    FROM GROWSCROPS G
    GROUP BY G.CROP
    HAVING COUNT(*)>1
);

```

```

/*

```

CROP	REGIONID	REGNAME
carrots	1	SWO
corn	1	SWO

```

*/

```

2 (ii). (5 marks) distributed as: 2.5 marks for the script file showing correct interaction with Oracle Sqlplus posing these queries; and 2.5 marks for correctly posing the queries and retrieving correct results.

3. Write four SQL update statements to do the following updates on the database schema shown in Figure 1.2. Show the affected tables after update through script file in sqlplus and in a script file created as before and named username_assn3que3.txt. (5 marks)

(Total for que 3 is 5 marks)

(a) Insert a new crop grown by an existing farmer in a region with unknown revenue as <10, 1, 'broccoli', 'NULL'> in the database.

(b) Change the crop budget of region with regname 'SWO' to 20.

(c) Insert a new farmer <60, 'Mary Tama', 21, 2>

(d) Delete all crops grown records for farmer whose SSN is 20.

Solution 3 (i): (5 marks)

```

-- 3 (a) Insert a new crop grown by an existing farmer in a region with unknown revenue as
<10, 1, 'broccoli', 'NULL'> in the database.

```

```

INSERT INTO GROWSCROPS values (10, 1, 'broccoli', 'NULL');

```

```

-- 1 row created.

```

```

-- 3 (b) Change the crop budget of region with regname 'SWO' to 20.

```

```

UPDATE REGION SET CROPBUDGET = 20 WHERE REGNAME = 'SWO';

```

```

-- 1 row updated.

```

```

-- 3. (c) Insert a new farmer <60, 'Mary Tama', 21, 2>

```

```

INSERT INTO FARMER VALUES (60, 'Mary Tama', 21, 2);

-- 1 row created.

-- 3 (d) Delete all crops grown records for farmer whose SSN is 20. .

DELETE FROM GROWSCROPS WHERE SSN = 20;

```

CHAPTER 7: More SQL: Complex Queries, Triggers, Views, and Schema Modification

4. (i) Write the following 2 queries in SQL on the database schema of Figure 1.2 using EXISTS or NOT EXISTS as appropriate.

(2.5 marks)

(ii) Implement the answering of your 2 queries in 4(i) using Sqlplus and the same database you created in question 1 and modified in earlier question with updates, deletes and inserts, providing your execution and answers to this question in a script file called username_assn3que4.

(2.5 marks)

(Total for que 4 is 5 marks)

(a) Retrieve the crop name and region id of all crops that have been grown in at least 2 regions.

(b) Retrieve the crop name and region id of all crops that have not been grown in at least 2 regions.

Solution 4 (i): (2.5 marks)

GA: check student queries as my answers are with the original database. A slightly different answer with correct queries from students should be accepted to accommodate use of modified database after implementing question 3 above.

```

-- 4 (a) Retrieve the crop name and region id of all crops that have been grown in at least
2 regions.

```

```

SELECT G.CROP, G.REGIONID
FROM GROWSCROPS G
WHERE EXISTS (
    SELECT CROP, count(G.Regionid)
    FROM GROWSCROPS
    WHERE G.CROP = CROP
    GROUP BY CROP
    HAVING COUNT(*) >=2
);

```

```

/*

```

CROP	REGIONID
potatoes	1
potatoes	2
peas	3
peas	3

```

potatoes                                4

*/

-- 4 (b) Retrieve the crop name and region id of all crops that have not been grown in at
least 2 regions.

SELECT G.CROP, G.REGIONID
FROM GROWSCROPS G
WHERE NOT EXISTS(
    SELECT CROP, count(G.Regionid)
    FROM GROWSCROPS
    WHERE G.CROP = CROP
    GROUP BY CROP
    HAVING COUNT(*) >=2
);

/*

CROP                                REGIONID
-----
carrots                            1
corn                                1

*/

```

4 (ii). (2.5 marks) distributed as: 0.5 marks for the script file showing correct interaction with Oracle Sqlplus posing these queries; and 2 marks for the correctly posing the queries and retrieving correct results.

5. In SQL, specify the following 3 queries on the COMPANY database of Figures 5.5 and 5.6 using the concept of nested queries and the concepts described in chapter 7. Note that you can just provide the SQL of these queries without building the database or implementing through Sqlplus. (Total for que 5 is 5 marks)

- Retrieve the names of all projects that have the maximum number of hours worked on them per week.
- Retrieve the hours of all projects whose number of hours worked on per week are greater than the average number of hours worked on all projects.
- Retrieve the names of projects that are worked on at least 20 hours more than the project with the least number of hours worked on it per week.

Figure 5.5 Schema diagram for the COMPANY relational database

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
-------	-------	-------	------------	-------	---------	-----	--------	-----------	-----

DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
-------	----------------	---------	----------------

DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
----------------	------------------

PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
-------	----------------	-----------	------

WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
-------------	------------	-------

DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
-------------	-----------------------	-----	-------	--------------

Solution 5: (5 marks)

- SELECT PNAME FROM PROJECT WHERE PNUMBER IN
(SELECT PNO FROM WORKS_ON WHERE HOURS IN
(SELECT MAX(HOURS) FROM WORKS_ON));
- SELECT HOURS FROM WORKS_ON WHERE HOURS >=ANY
(SELECT AVG(HOURS) FROM WORKS_ON);
- SELECT PNO, HOURS FROM WORKS_ON WHERE HOURS >= 20 +
(SELECT MIN(HOURS) FROM WORKS_ON);

CHAPTER 8: THE RELATIONAL ALGEBRA AND RELATIONAL CALCULUS

6. Specify the following 5 queries on the COMPANY relational database schema shown in Figure 5.5, using the relational operators discussed in chapter 8. Also show the result of each query as it would apply to the database state of Figure 5.6. (Total for que 6 is 10 marks)

Some symbols for solving queries you may copy and reuse are: π , σ , $\bowtie$, ρ , $\Join$

Figure 5.6 One possible database state for the COMPANY relational database schema.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

- (i) List the names of employees who have a dependent.
- (ii) Find the names of employees that are directly supervised by 'Jennifer Wallace'.
- (iii) For each project, list the project name and the average hours per week (by all employees) spent on that project.
- (iv) Retrieve the names of employees who work on every project.
- (v) Retrieve the minimum salary of all male employees.

Solution 6: (10 marks)

In the relational algebra, as in other languages, it is possible to specify the same query in multiple ways. We give one possible solution for each query. We use the symbol σ for SELECT, ρ for PROJECT, $\Join$ for EQUIJOIN, $*$ for NATURAL JOIN, and f for FUNCTION.

(i)

$$E \leftarrow (\text{EMPLOYEE} \Join_{\text{Ssn}=\text{Essn}} (\text{DEPENDENT}))$$

$$R \leftarrow \pi_{\text{Lname}, \text{Fname}} (E)$$

Result (empty):

LNAME	FNAME
Wong	Franklin
Wallace	Jennifer
Smith	John

(ii)) $\text{Wallace_ssn} \leftarrow \pi_{\text{Ssn}} (\sigma_{\text{Fname}='Jennifer' \text{ and } \text{Lname}='Wallace'} (\text{EMPLOYEE}))$
 $\text{Wallace_emps} \leftarrow (\text{EMPLOYEE} \Join_{\text{Super_ssn}=\text{Ssn}} (\text{Wallace_ssn}))$
 $\text{Result} \leftarrow \pi_{\text{Lname}, \text{Fname}} (\text{Wallace_emps})$

Result:

Select e.fname, e.lname
 From employee e, employee m
 Where e.super_ssn = m.ssn and m.fname = 'Jennifer' and m.lname='Wallace';

FNAME	LNAME
Ahmad	Jabbar

(iii)

$$\text{PROJ_HOURS} (\text{Pno}, \text{Avg_Hrs}) \leftarrow \text{Pno} \Join_{\text{Avg_Hours}} (\text{WORKS_ON})$$

$$\text{RESULT} \leftarrow \pi_{\text{Pname}, \text{Avg_hrs}} (\text{PROJ_HOURS} \Join_{\text{Pno}=\text{Pnumber}} (\text{PROJECT}))$$

select pname, avg(hours)
 from works_on, project
 where pno = pnumber
 group by pname;

Result:

PNAME	AVG(HOURS)
Nbenefits	30
Computerize	18.33
ProductZ	25
Reorganize	10
ProductY	12.5

ProductX 26.25

(iv)

PROJ_EMPS(PNO,SSN) <-- $\pi_{pno, Essn}$ (WORKS_ON)

ALL_PROJS(PNO) <-- $\pi_{PNUMBER}$ (PROJECT)

EMPS_ALL_PROJS <-- PROJ_EMPS :- ALLPROJS /* DIVISION operation */

RESULT <-- $\pi_{LNAME,FNAME}$ (EMPLOYEE $\bowtie$ EMP_ALL_PROJS) /*natural join on ssn*/

Result (empty):

LNAME FNAME

(v) RESULT(MAX_F_SAL) $\leftarrow \mathfrak{I}_{\text{minimum salary}} ((\sigma_{SEX = 'M'} \text{ EMPLOYEE}))$

Or

(v) ρ RESULT(MAX-F-SAL) ($\mathfrak{I}_{\text{minimum salary}} (\sigma_{SEX = 'M'} \text{ EMPLOYEE}))$

select min(salary)

from employee

where sex = 'M';

Result:

MIN(SALARY)

25000